AdaBoost Does Not Always Cycle: A Computer-Assisted Counterexample

Erik Y. Wang*

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Abstract

We give a computer-assisted counterexample to the open question, posed by Rudin, Schapire, and Daubechies in COLT 2012, of whether exhaustive AdaBoost always converges to a finite cycle. The construction is based on a block-product gadget whose two factors share an exact period-2 orbit for their 5-step branch maps, but whose linearized return maps have dominant eigenvalues with an irrational logarithmic ratio. This irrationality forces the burst-winner sequence to have an irrational asymptotic frequency, precluding eventual periodicity. All assertions are certified by exact rational arithmetic. This work was developed in collaboration with GPT-5.4 Pro and Claude Opus 4.6.

1 Introduction

The AdaBoost algorithm [2] defines a discrete dynamical system on the probability simplex in $\mathbb{R}^m$ [6], where a sequence of distributions D_t is updated based on an $m \times N$ matrix $M \in \{-1, +1\}^{m \times N}$ representing the correlations between N weak hypotheses and m training examples. Under the exhaustive selection of weak hypotheses, the next distribution is determined by the mapping $D_{t+1}(i) = D_t(i)/(1 + r_t M_{ij_t})$, where $j_t = \operatorname{argmax}_j (D_t^\top M)_j$ and $r_t = (D_t^\top M)_{j_t}$. While it is proven that D_t converges to a unique fixed point when the weak learning assumption ($r_t \geq c > 0$) is violated, the behavior remains mathematically elusive when the assumption holds. The iterative map is highly discontinuous due to the argmax operation, leaving the long-term asymptotic behavior—specifically whether the sequence $\{D_t\}$ always enters a cycle of length l such that $D_{kl+b} \to \bar{D}_b$ as $k \to \infty$—an open question [6, 7].

The problem under consideration is the following: prove or disprove that for every $\{-1, +1\}$-valued matrix M, the sequence of distributions D_t computed by the AdaBoost [7] mapping necessarily converges to a cycle. In this paper, we disprove the conjecture by constructing an explicit counterexample.[1]

Proof outline

We construct a 20×8 sign matrix $M_0 = L_A \boxplus L_B$ as a block product of two small gadgets whose dynamics decouple on an invariant product manifold (Section 3). Both gadgets share an exact period-2 orbit for the 5-step branch maps (Section 6). The linearized return maps at this orbit have dominant eigenvalues λ and κ whose ratio $\log \lambda / \log \kappa$ is irrational, as shown by a norm argument in

*Department of Computer Science, Stanford University. Email: `erikwang@stanford.edu`

[1]The computational certificate is available at this link: `https://github.com/ewang26/AdaBoost`

algebraic number theory (Section 7). Rigorous interval arithmetic certifies explicit invariant sectors around the orbit, together with contraction rates and burst-locking margins (Sections 8–10). Within these sectors, the irrational eigenvalue ratio forces the burst-winner sequence to have an irrational asymptotic frequency, which precludes eventual periodicity and hence finite-cycle convergence (Sections 11–13). A row-duplication argument converts an explicit rational starting point into a uniform-start counterexample on a larger matrix $\widetilde{M}$ with weak-learning margin 1/5 (Section 14).

2 Main result

We work with exhaustive AdaBoost on a fixed sign matrix $M \in \{-1,+1\}^{m\times N}$, with smallest-index tie-breaking. Defining the *edge* $\mu_t(j)$ of hypothesis j at time t by

$$\mu_t(j) = (D_t^\top M)_j, \qquad j_t = \min \operatorname*{argmax}_{1\le j\le N} \mu_t(j), \qquad r_t = \mu_t(j_t),$$

the update is

$$D_{t+1}(i) = \frac{D_t(i)}{1 + r_t M_{ij_t}}.$$

We call the distribution $D_t \in \Delta^{m-1}$ the *state* of the system at time t; the map $D_t \mapsto D_{t+1}$ defined above is a single AdaBoost *update step*. Let Δ^{m-1} be the $m-1$-dimensional simplex in $\mathbb{R}^m$. The *weak-learning margin* of M is

$$\gamma(M) = \min_{D\in\Delta^{m-1}} \max_{1\le j\le N} (D^\top M)_j.$$

We show the following result:

Theorem 1. *There exists a finite $\{-1,+1\}$-valued matrix $\widetilde{M}$ with weak-learning margin $\gamma(\widetilde{M}) = \frac{1}{5} > 0$ such that standard exhaustive AdaBoost on $\widetilde{M}$, started from the exact uniform distribution and using smallest-index tie-breaking, does **not** converge to any finite cycle.*

*Consequently, the COLT question "does exhaustive AdaBoost always converge to a finite cycle?" has a **negative** answer.*

Corollary 2. *There exists a finite $\{-1,+1\}$-valued matrix $\widetilde{M}'$ whose column set is closed under negation (i.e., for every column h of $\widetilde{M}'$, $-h$ also appears as a column of $\widetilde{M}'$), with weak-learning margin $\gamma(\widetilde{M}') = \frac{1}{5} > 0$, such that standard exhaustive AdaBoost on $\widetilde{M}'$, started from the exact uniform distribution and using smallest-index tie-breaking, does **not** converge to any finite cycle. In particular, the COLT question in its original hypothesis-class-closed-under-negation formulation also has a negative answer.*

The proof of the Corollary appears at the end of Section 14.

The remainder of the paper is devoted to the proof of Theorem 1, which occupies Sections 3–14.

Our work comes with an accompanying symbolic certificate using only symbolic algebra over $\mathbb{Q}$ (via [4]), exact interval arithmetic, and exact square-root enclosures obtained by integer arithmetic. In the presentation of this paper, decimal intervals are used only for reporting.

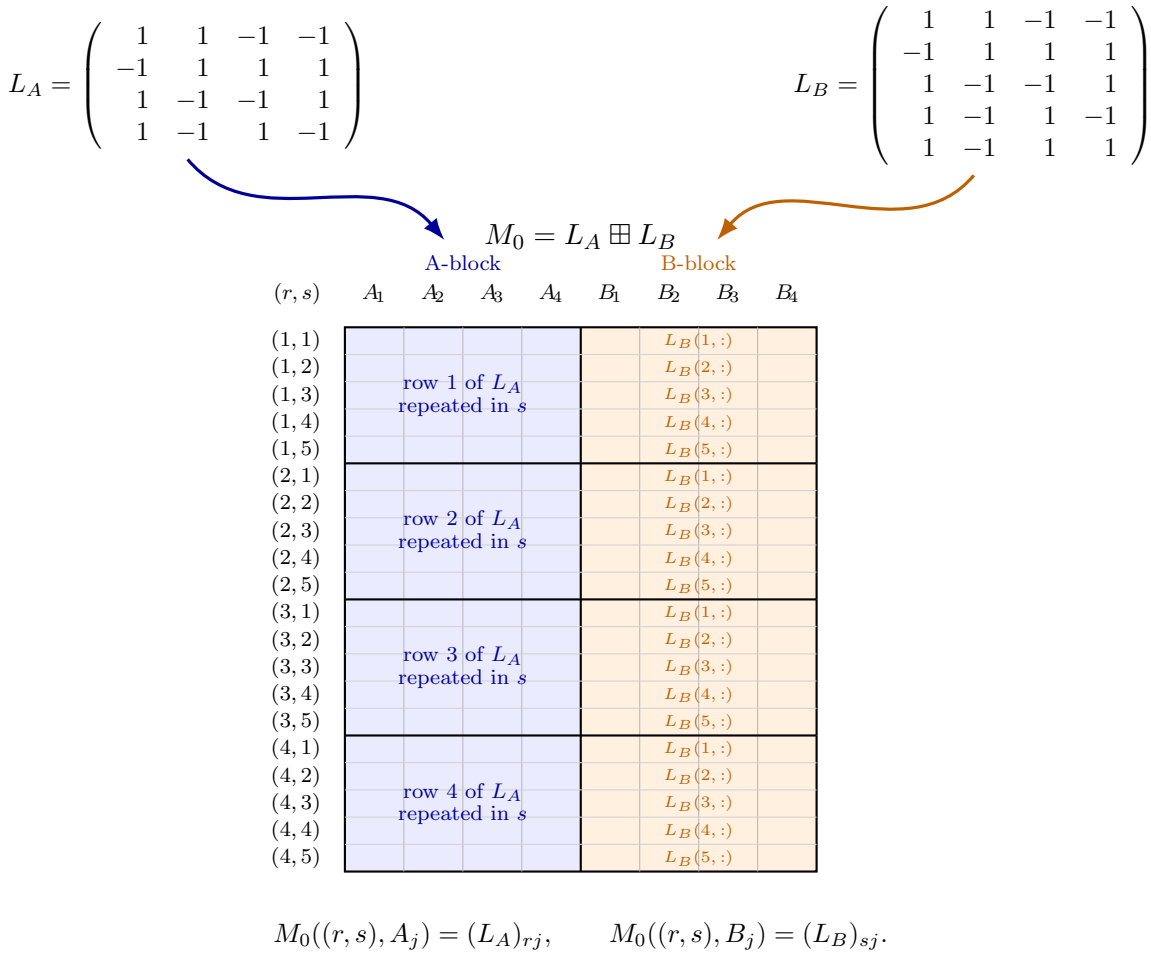


Figure 1: Schematic construction of $M_0 = L_A \boxplus L_B$. Rows are indexed by $(r,s) \in \{1,2,3,4\} \times \{1,2,3,4,5\}$. The first four columns are copied from row r of L_A, while the last four columns are copied from row s of L_B.

3 The gadgets, the product system, and the exact invariant affine manifolds

The counterexample is built from two small sign matrices, which we call *gadgets*. Let

$$L_A = \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad L_B = \begin{pmatrix} 1 & 1 & -1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix}.$$

Define the 20×8 block-product matrix $M_0 = L_A \boxplus L_B$ as follows (see Figure 1). Its rows are indexed by pairs $(r,s) \in \mathcal{I} := \{1,2,3,4\} \times \{1,2,3,4,5\}$, and its columns are denoted $A_1, \ldots, A_4, B_1, \ldots, B_4$. The entries are

$$M_0\big((r,s), A_j\big) = (L_A)_{rj}, \qquad M_0\big((r,s), B_j\big) = (L_B)_{sj}.$$

Lemma 3 (Reduction to row marginals)**.** *Let $W = (w_{rs})_{(r,s)\in\mathcal{I}}$ be any distribution on the* 20 *rows of M_0, and define its row marginals*

$$a_r = \sum_{s=1}^{5} w_{rs}, \qquad b_s = \sum_{r=1}^{4} w_{rs}.$$

Then for the edge vector $W^\top M_0 \in \mathbb{R}^8$*, whose jth component $(W^\top M_0)_j = \sum_{(r,s)} w_{rs}(M_0)_{(r,s),j}$ is the inner product of W with column j, the A-block edges depend only on a, and the B-block edges depend only on b:*

$$\begin{aligned}(W^\top M_0)_{A_j} &= (a^\top L_A)_j \qquad (1 \le j \le 4),\\ (W^\top M_0)_{B_j} &= (b^\top L_B)_j \qquad (1 \le j \le 4).\end{aligned}$$

Proof. For $1 \le j \le 4$,

$$(W^\top M_0)_{A_j} = \sum_{r=1}^{4}\sum_{s=1}^{5} w_{rs}(L_A)_{rj} = \sum_{r=1}^{4} a_r(L_A)_{rj} = (a^\top L_A)_j.$$

The B-block identity is similar. □

Lemma 4 (Exact invariance of the product manifold)**.** *Let*

$$\mathcal{P} = \{a \otimes b : \ a \in \Delta^3, \ b \in \Delta^4\}.$$

If $W = a \otimes b \in \mathcal{P}$ and AdaBoost selects an A-column A_j, then the updated state is $W' = a' \otimes b$, where a' is the one-step exhaustive-AdaBoost update of a on L_A using column j. If AdaBoost selects a B-column B_j, then $W' = a \otimes b'$, where b' is the one-step update of b on L_B using column j. Hence $\mathcal{P}$ is invariant *under the AdaBoost dynamics on M_0: if $W \in \mathcal{P}$, then the updated state $W' \in \mathcal{P}$.*

Proof. Assume an A-column j is selected. By Lemma 3, the selected edge is $\mu = (a^\top L_A)_j$. Therefore

$$w'_{rs} = \frac{a_r b_s}{1 + \mu \cdot (L_A)_{rj}} = \left(\frac{a_r}{1 + \mu \cdot (L_A)_{rj}}\right) b_s = a'_r b_s,$$

where a' is the one-step AdaBoost update of a on L_A. The B-column case is identical. □

We will show in Section 10 that the product dynamics over $\mathcal{P}$ decouples into alternating blocks of five consecutive AdaBoost steps in which the same gadget is selected at every step. We call each such block of five steps a *burst*. Within a burst, the column-selection sequence on the active gadget is one of two possible 5-tuples, called *half-words*,

$$H_0 = (1, 3, 4, 1, 2), \qquad H_1 = (1, 4, 3, 1, 2).$$

The choice between these is determined by the *phase*: if the active gadget's current state is near the first point $\hat{P}_0^G$ of a certain 2-cycle (Section 6), we say it is in phase 0; if near $\hat{P}_1^G$, in phase 1.

Since both H_0 and H_1 end in column 2, and column 2 of L_A is $(1, 1, -1, -1)^\top$, the classical balancing lemma for AdaBoost (see [8] and Section A.1) implies that after the final step of any A-burst, the total mass on the $+1$ rows (rows $1, 2$) and on the -1 rows (rows $3, 4$) is each exactly $\frac{1}{2}$. Writing the distribution as $(a, b, c, d) \in \Delta^3$, this means $a + b = \frac{1}{2}$ and $c + d = \frac{1}{2}$. Hence every 5-step A-burst lands in

$$\Sigma_A = \{(a, b, c, d) \in \Delta^3 : \ a + b = \tfrac{1}{2}, \ c + d = \tfrac{1}{2}\}.$$

Similarly, column 2 of L_B is $(1,1,-1,-1,-1)^\top$, so Lemma 10 gives $a+b=\frac{1}{2}$ and $c+d+e=\frac{1}{2}$. Every 5-step B-burst lands in

$$\Sigma_B = \{(a,b,c,d,e) \in \Delta^4 : \ a+b=\tfrac{1}{2},\ c+d+e=\tfrac{1}{2}\}.$$

Thus, for some p, d, we can represent

$$(a,b,c,d) = \Big(p-\tfrac{1}{2},\ 1-p,\ \tfrac{1}{2}-d,\ d\Big) \in \Sigma_A,$$
$$(a,b,c,d,e) = \Big(p-\tfrac{1}{2},\ 1-p,\ \tfrac{1}{2}-d-e,\ d,\ e\Big) \in \Sigma_B.$$

For each gadget $G \in \{A,B\}$ and each half-word H_i, composing the five consecutive AdaBoost update steps prescribed by H_i yields a rational map $\Sigma_G \to \Sigma_G$ called the *branch map* F_i^G (explicit formulas are given in Section 5). Item 2 of the certificate (Appendix A.2) verifies symbolically that each numerator polynomial of the last coordinate of F_i^B is divisible, as a polynomial in $\mathbb{Q}[p,d,e]$, by the variable e. Consequently, the *face* $\{e=0\} \subset \Sigma_B$—i.e., the subset of Σ_B obtained by setting the fifth simplicial coordinate to zero—is invariant under both branch maps. By inspection, restriction to this face recovers the A-gadget maps:

$$F_0^B|_{e=0} = F_0^A, \qquad F_1^B|_{e=0} = F_1^A.$$

Notation convention. Throughout, we distinguish *affine chart points* from *simplex probability vectors* using a bar. We write

$$\hat{P}_i^A = (p_i, d_i) \in \mathbb{R}^2, \qquad \hat{P}_i^B = (p_i, d_i, e_i) \in \mathbb{R}^3$$

for chart coordinates on Σ_A and Σ_B, and $P_i^A \in \Delta^3$, $P_i^B \in \Delta^4$ for the corresponding simplex probability vectors obtained from the affine parametrizations above. Similarly, at burst boundary n (i.e., the state immediately before the nth burst begins), we write $\hat{a}_n, \hat{b}_n$ for chart coordinates and a_n, b_n for the simplex vectors. We define $W_n = a_n \otimes b_n \in \mathcal{P}$ to be the product distribution on the 20 rows of M_0 at the nth burst boundary.

4 Margin

We show using a single linear witness that the weak-learning margin of the block-product matrix M_0 is exactly $1/5$. Let $\beta = (2,1,1,1)^\top \in \mathbb{R}^{1\times 4}$. Every row r of L_A satisfies $r\beta = 1$, and every row r of L_B satisfies $r\beta \in \{1,3\}$. For any gadget $G \in \{A,B\}$ and any distribution u on the rows of L_G, write $\mu_j = (u^\top L_G)_j$ for the edge of column j. Then

$$2\mu_1 + \mu_2 + \mu_3 + \mu_4 = u^\top L_G\, \beta \geq 1,$$

so $\max_j \mu_j \geq 1/5$. Hence $\gamma(L_A), \gamma(L_B) \geq \frac{1}{5}$.

Equality for L_A is attained at

$$u_* = (1/5, 2/5, 1/5, 1/5), \qquad u_*^\top L_A = (1/5, 1/5, 1/5, 1/5).$$

It follows that $\gamma(L_A) = 1/5$. Since the first four rows of L_B coincide with L_A, the same argument gives $\gamma(L_B) \leq 1/5$, and together with the lower bound above, $\gamma(L_A) = \gamma(L_B) = \frac{1}{5}$.

By Lemma 3, for any distribution W on the rows of M_0 with marginals (a,b),

$$\max_{1\leq j\leq 8} (W^\top M_0)_j = \max\Big(\max_{1\leq j\leq 4} (a^\top L_A)_j,\ \max_{1\leq j\leq 4} (b^\top L_B)_j\Big).$$

Conversely, every pair of marginals (a,b) is realized by the product distribution $a \otimes b$. Hence

$$\gamma(M_0) = \min_W \max_j (W^\top M_0)_j = \min_{a,b} \max\Big(\max_j (a^\top L_A)_j,\ \max_j (b^\top L_B)_j\Big) = \frac{1}{5}.$$

5 Exact 5-step branch maps

We now compute the closed-form rational maps that result from composing five consecutive AdaBoost updates along each half-word. The two branch maps F_0^G and F_1^G for each gadget $G \in \{A, B\}$ are related by a simple involution. Define the *phase-swap* maps

$$S_A(p, d) = \big(p, \tfrac{1}{2} - d\big), \qquad S_B(p, d, e) = \big(p, \tfrac{1}{2} - d - e,\ e\big).$$

Each S_G is an involution ($S_G^2 = \mathrm{id}$). Using simple algebra, one can check that in simplex coordinates, both maps S_G *transpose the 3rd and 4th distribution entries* while leaving the rest fixed.

For either gadget, rows 3 and 4 of L_G differ only in columns 3 and 4 (where their signs are opposite), so this row transposition induces the column transposition $\pi = (3\ 4)$ on the edge vector. If T_j^G denotes the one-step AdaBoost update using column j of L_G, then $S_G \circ T_j^G \circ S_G = T_{\pi(j)}^G$. Because $S_G^2 = \mathrm{id}$, conjugating the five-step composition F_0^G by S_G replaces each column index j with $\pi(j)$, transforming $H_0 = (1, 3, 4, 1, 2)$ into $H_1 = (1, 4, 3, 1, 2)$. Hence,

$$F_1^A = S_A \circ F_0^A \circ S_A, \qquad F_1^B = S_B \circ F_0^B \circ S_B.$$

It therefore suffices to give explicit formulas for F_0^A and F_0^B.

A-gadget

To obtain F_0^A, we substitute the chart parametrization $(p - \frac{1}{2}, 1 - p, \frac{1}{2} - d, d) \in \Sigma_A$ into the five successive AdaBoost update steps $a_i \mapsto a_i/(1 + \mu\, h_i)$ for the columns of L_A prescribed by $H_0 = (1, 3, 4, 1, 2)$, then express the final result in the chart coordinates (p, d) and reduce to lowest terms. The phase-0 branch map on Σ_A is the rational function F_0^A such that for all chart coordinates p, d,

$$F_0^A(p, d) = \left(\frac{P_A(p, d)}{Q_A(p, d)}, \frac{D_A(p, d)}{R_A(p, d)}\right),$$

where

$$P_A = 4d^4 - 6d^3p + 2d^3 - 30d^2p^2 + 15d^2p - 2d^2 - 6dp^3 + 14dp^2 - 4dp + 6p^4 + p^3 - 2p^2,$$

$$Q_A = 8d^4 - 8d^3p + 2d^3 - 48d^2p^2 + 22d^2p - 3d^2 - 8dp^3 + 22dp^2 - 6dp + 8p^4 + 2p^3 - 3p^2,$$

$$D_A = d\big(2d^3 + 2d^2p - d^2 - 6dp^2 + 2p^3 + p^2\big),$$

and

$$R_A = 8d^4 - 2d^3 - 24d^2p^2 + 6d^2p - d^2 + 10dp^2 - 2dp + 2p^3 - p^2.$$

B-gadget

The same five-step composition applied to L_B along H_0, in the chart coordinates (p, d, e) on Σ_B, gives the phase-0 branch map on Σ_B as

$$F_0^B(p, d, e) = \left(\frac{P_B(p, d, e)}{Q_B(p, d, e)}, \frac{D_B(p, d, e)}{R_B(p, d, e)}, \frac{E_B(p, d, e)}{R_B(p, d, e)}\right),$$

where

$$\begin{aligned}P_B = {} & 4d^4 + 16d^3e - 6d^3p + 2d^3 + 20d^2e^2 - 36d^2ep + 12d^2e - 30d^2p^2 + 15d^2p - 2d^2 \\ & + 8de^3 - 46de^2p + 18de^2 - 48dep^2 + 30dep - 4de - 6dp^3 + 14dp^2 - 4dp \\ & - 16e^3p + 8e^3 - 22e^2p^2 + 15e^2p - 2e^2 + 8ep^2 - 4ep + 6p^4 + p^3 - 2p^2,\end{aligned}$$

$$\begin{aligned}Q_B = {} & 8d^4 + 32d^3e - 8d^3p + 2d^3 + 40d^2e^2 - 56d^2ep + 16d^2e - 48d^2p^2 + 22d^2p - 3d^2 \\ & + 16de^3 - 72de^2p + 26de^2 - 72dep^2 + 44dep - 6de - 8dp^3 + 22dp^2 - 6dp \\ & - 24e^3p + 12e^3 - 32e^2p^2 + 22e^2p - 3e^2 + 12ep^2 - 6ep + 8p^4 + 2p^3 - 3p^2,\end{aligned}$$

$$D_B = d\big(2d^3 + 8d^2e + 2d^2p - d^2 + 10de^2 - 2de - 6dp^2 + 4e^3 - 2e^2p - e^2 - 4ep^2 + 2p^3 + p^2\big),$$

$$\begin{aligned}R_B = {} & 8d^4 + 32d^3e - 2d^3 + 40d^2e^2 - 24d^2ep - 24d^2p^2 + 6d^2p - d^2 \\ & + 16de^3 - 32de^2p + 6de^2 - 24dep^2 + 12dep - 2de + 10dp^2 - 2dp \\ & - 8e^3p + 4e^3 - 8e^2p^2 + 6e^2p - e^2 + 4ep^2 - 2ep + 2p^3 - p^2,\end{aligned}$$

and

$$E_B = e\big(2d^3 + 4d^2e - 6d^2p + d^2 + 2de^2 - 8dep + 2de + 2dp^2 - 2e^2p + e^2 + 2p^3 - p^2\big).$$

Item 1 of the certificate (Appendix A.2) verifies symbolically that the displayed polynomials for both gadgets match the five-step compositions term-for-term. Because E_B has a factor e, the face $e = 0$ is invariant.

Lemma 5 (Regularity of the branch maps)**.** *The denominators Q_A, R_A are nonzero on $\Sigma_A^\circ := \{(p,d) : \frac{1}{2} < p < 1,\ 0 < d < \frac{1}{2}\}$, and Q_B, R_B are nonzero on $\Sigma_B^\circ = \{(p,d,e) : \frac{1}{2} < p < 1,\ d > 0,\ e > 0,\ d + e < \frac{1}{2}\}$. Moreover, the branch maps F_i^A, F_i^B and all their derivatives are smooth on these domains.*

Proof. Observe that every column of L_A contains both $+1$ and -1 entries.

Strict positivity implies $|\mu| < 1$. If $a \in \Delta^{m-1}$ is strictly positive and $h \in \{-1, +1\}^m$ has both signs, then

$$\mu = a^\top h = \sum_{h_i=+1} a_i - \sum_{h_i=-1} a_i.$$

Both partial sums are strictly positive, so $-1 < \mu < 1$. Hence every denominator $1 + \mu h_i \in \{1 + \mu,\ 1 - \mu\}$ is strictly positive, and the updated vector $a_i' = a_i/(1 + \mu h_i)$ is again strictly positive with $\sum_i a_i' = 1$.

Well-definedness of the five-step composition. A point $(p,d) \in \Sigma_A^\circ$ corresponds to the strictly positive distribution $(p - \frac{1}{2},\ 1 - p,\ \frac{1}{2} - d,\ d) \in \Delta^3$. By the above reasoning, each of the five individual AdaBoost updates in the column sequence H_0 (or H_1) has strictly positive denominators and produces a strictly positive output distribution. Hence the five-step composition is well-defined on all of Σ_A°. An identical argument applies to L_B on Σ_B°.

Reduced denominators are nonzero. Before algebraic simplification, each coordinate of the five-step composition is a rational function $\tilde{P}/\tilde{Q}$ whose denominator $\tilde{Q}$ is a product of the individual-step denominators (polynomials in the chart coordinates obtained by successive substitution and clearing of fractions). By the above reasoning, each factor is strictly positive on Σ_A°, so $\tilde{Q} > 0$ there. The reduced-form denominator Q satisfies $Q \mid \tilde{Q}$: since $P\tilde{Q} = Q\tilde{P}$ and $\gcd(P, Q) = 1$, Euclid's lemma in the polynomial ring $\mathbb{Q}[p,d]$ gives $Q \mid \tilde{Q}$. Hence any real zero of Q would also be a zero of $\tilde{Q}$. Since $\tilde{Q}$ has no zeros on Σ_A°, neither does Q. The same argument applies to R_A, Q_B, and R_B.

Since F_i^A and F_i^B are rational functions with nonzero denominators, they are C^∞ on Σ_A° and Σ_B° respectively; finishing the proof. □

6 Exact common 2-cycle

We now argue that both gadgets share an exact period-2 orbit for the 5-step branch maps. By the conjugacy $F_1^A = S_A \circ F_0^A \circ S_A$ (Section 5), if F_0^A maps a point (p, d) to its phase-swap $S_A(p, d) = (p, \frac{1}{2} - d)$, then F_1^A maps the phase-swap back: $F_1^A(p, \frac{1}{2} - d) = S_A(F_0^A(p, d)) = (p, d)$, giving a 2-cycle. We therefore seek solutions of

$$F_0^A(p, d) = \big(p, \tfrac{1}{2} - d\big).$$

Substituting the explicit rational formulas for F_0^A (Section 5), the two coordinate equations $P_A/Q_A = p$ and $D_A/R_A = \frac{1}{2} - d$ become, after cross-multiplying and removing factors that are nonzero on Σ_A°, the polynomial system $N_1(p, d) = 0$, $N_2(p, d) = 0$, where

$$N_1(p, d) = 2d^4 - 2d^3p + d^3 - 12d^2p^2 + 7d^2p - d^2 - 2dp^3 + 6dp^2 - 2dp + 2p^4 - p^2,$$

and

$$\begin{aligned} N_2(p, d) = & -16d^5 + 8d^4 + 48d^3p^2 - 16d^3p + 2d^3 \\ & - 32d^2p^2 + 10d^2p - d^2 - 8dp^3 + 10dp^2 - 2dp + 2p^3 - p^2. \end{aligned}$$

To eliminate d from the system $N_1 = N_2 = 0$, we compute the resultant of N_1 and N_2 with respect to d. By standard elimination theory [1], any common root (p, d) must satisfy $\mathrm{Res}_d(N_1, N_2)(p) = 0$. Item 3 of the certificate (Appendix A.2) verifies that

$$\mathrm{Res}_d(N_1, N_2) = 4096\, p^{10}(2p - 1)^3 f(p),$$

where

$$f(p) = 1024p^7 - 3072p^6 + 3200p^5 - 1152p^4 - 144p^3 + 176p^2 - 40p + 7.$$

The factors $p^{10}(2p - 1)^3$ are nonzero on Σ_A° (since $p > \frac{1}{2}$), so any interior solution must satisfy $f(p) = 0$.

The certificate (Item 3 of Appendix A.2) isolates a unique root

$$p_* \in [0.637116837818846,\ 0.637116837818848],$$

and proves

$$\begin{aligned} d_* &= 512p_*^6 - 1152p_*^5 + 736p_*^4 - 24p_*^3 - 88p_*^2 + 19p_* - \tfrac{9}{2} \\ &\in [0.257995382953758869,\ 0.257995382958145447]. \end{aligned}$$

In chart coordinates, set

$$\hat{P}_0^A = (p_*,\, d_*), \qquad \hat{P}_1^A = (p_*,\, \tfrac{1}{2} - d_*).$$

The corresponding simplex vectors are

$$P_0^A = \Big(p_* - \tfrac{1}{2},\ 1 - p_*,\ \tfrac{1}{2} - d_*,\ d_*\Big), \qquad P_1^A = \Big(p_* - \tfrac{1}{2},\ 1 - p_*,\ d_*,\ \tfrac{1}{2} - d_*\Big). \tag{6.1}$$

Then $F_0^A(\hat{P}_0^A) = \hat{P}_1^A$, $F_1^A(\hat{P}_1^A) = \hat{P}_0^A$. In chart coordinates, set

$$\hat{P}_0^B = (p_*,\, d_*,\, 0), \qquad \hat{P}_1^B = (p_*,\, \tfrac{1}{2} - d_*,\, 0),$$

with corresponding simplex vectors

$$P_0^B = (P_0^A, 0), \qquad P_1^B = (P_1^A, 0). \tag{6.2}$$

Since $F_i^B|_{e=0} = F_i^A$, the same identities hold for B: $F_0^B(\hat{P}_0^B) = \hat{P}_1^B$, $F_1^B(\hat{P}_1^B) = \hat{P}_0^B$.

Thus both gadgets share an exact period-2 orbit for the 5-step branch maps, equivalently a period-10 orbit in AdaBoost time. We say gadget G is in *phase* i ($i \in \{0, 1\}$) when its state lies in the sector S_i^G (Section 9); within these sectors, the column-selection sequence is exactly H_i (Section 10), so the 5-step composition is the branch map F_i^G.

7 Exact spectra and the irrationality input

Denote by $J_i^G = DF_i^G(\hat{P}_i^G)$ the Jacobian of the branch map F_i^G at the 2-cycle point $\hat{P}_i^G$, and define the *return maps*

$$R_A = J_1^A J_0^A, \qquad R_B = J_1^B J_0^B.$$

Each R_G is the linearization of one full period (phase $0 \to 1 \to 0$) of the 2-cycle for gadget G. We call the binary sequence recording which gadget (A or B) is selected at each successive burst the *burst-winner sequence* (defined precisely in Section 11).

The key algebraic input of this section is that the dominant eigenvalues of R_A and R_B have an irrational logarithmic ratio, which will ultimately force the burst-winner sequence to be aperiodic.

Next, observe that the return map R_B has a block-triangular structure:

$$R_B = \begin{pmatrix} R_A & u \\ 0 & \kappa \end{pmatrix}.$$

This block-triangular structure holds because the e-component E_B of each branch map factors as $e \cdot g(p, d, e)$, so at $e = 0$ the partial derivatives $\partial E_B/\partial p$ and $\partial E_B/\partial d$ vanish. Each $J_i^B|_{e=0}$ therefore has zeros in positions $(3, 1)$ and $(3, 2)$, and matrix multiplication preserves this pattern. The upper-left 2×2 block equals R_A because $F_i^B|_{e=0} = F_i^A$.

The dominant eigenvalue of R_A is an algebraic number λ whose irreducible polynomial over $\mathbb{Q}$ is needed explicitly to compute the field norm for the irrationality argument below. It turns out that this value λ is the unique root of the irreducible degree-14 polynomial

$$\begin{aligned} P_\lambda(x) = {} & 16x^{14} - 117232x^{13} - 8330040x^{12} - 3628852593x^{11} \\ & + 766183444916x^{10} - 37440753370248x^9 \\ & + 504085851461616x^8 + 57085439654130x^7 \\ & + 18299281845941356x^6 - 5676298472505780x^5 \\ & + 422743039128068x^4 - 4789675432569x^3 \\ & + 7468158864x^2 - 2899292x + 4 \end{aligned}$$

in the *isolating interval* $[0.0944920594891271,\ 0.0944920594891272]$. The certificate derives P_λ as the elimination polynomial for λ and confirms it is irreducible (Item 5, Appendix A.2), so P_λ is the minimal polynomial of λ over $\mathbb{Q}$. The isolating interval contains exactly one real root of P_λ (Item 7), thereby identifying λ uniquely among the roots.

If α is an algebraic number with irreducible polynomial $a_d x^d + \cdots + a_0$ over $\mathbb{Q}$, its *field norm* $N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha)$ is the product of all roots of this polynomial, which equals $(-1)^d a_0/a_d$ by Vieta's formulas [3]. Hence

$$N_{\mathbb{Q}(\lambda)/\mathbb{Q}}(\lambda) = \frac{4}{16} = \frac{1}{4}.$$

The other eigenvalue of R_A is also a root of P_λ.

Similarly, the eigenvalue κ of R_B not inherited from R_A turns out to be the unique root of the irreducible degree-7 polynomial

$$P_\kappa(x) = x^7 - 35x^6 + 385x^5 - 1835x^4 + 2315x^3 - 625x^2 + 51x - 1$$

in the isolating interval $[0.2038817328642625,\ 0.2038817328642626]$. As above, the certificate derives P_κ as the irreducible elimination polynomial for κ (Item 6), making it the minimal polynomial of κ over $\mathbb{Q}$, and verifies that this interval contains exactly one real root (Item 7). Hence $N_{\mathbb{Q}(\kappa)/\mathbb{Q}}(\kappa) = 1$.

Defining ν as the unique root of P_λ in $[0.0011453882842312,\ 0.0011453882842313]$ (Item 7 of the certificate), we then find that the isolating intervals give $0 < \nu < \lambda < \kappa < 1$.

If $\log\lambda/\log\kappa \in \mathbb{Q}$, then $\lambda^m = \kappa^n$ for some integers $m, n > 0$. Let $L = \mathbb{Q}(\lambda, \kappa)$ be the *compositum*, the smallest number field containing both λ and κ. The field norm is multiplicative ($N(\alpha\beta) = N(\alpha)N(\beta)$) and satisfies the tower formula $N_{L/\mathbb{Q}}(\alpha) = N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha)^{[L:\mathbb{Q}(\alpha)]}$, where $[L : \mathbb{Q}(\alpha)]$ denotes the degree of the field extension $L/\mathbb{Q}(\alpha)$, again using Vieta's formulas [3]. Applying $N_{L/\mathbb{Q}}$ to both sides of $\lambda^m = \kappa^n$ gives

$$(1/4)^{m[L:\mathbb{Q}(\lambda)]} = 1^{n[L:\mathbb{Q}(\kappa)]},$$

which is impossible since the left side is a positive power of $1/4$. Therefore

$$\frac{\log\lambda}{\log\kappa} \notin \mathbb{Q}.$$

8 Exact dominant half-step directions and half-step contractions

The return map $R_G = J_1^G J_0^G$ is the Jacobian of the composition $F_1^G \circ F_0^G$ at the fixed point $\hat{P}_0^G$ (Section 7), and thus governs the first-order dynamics near the 2-cycle, but the proof requires a finer decomposition into half-steps. The phase symmetries from Section 5 relate the two Jacobians at corresponding phases of the 2-cycle. Recall that $F_1^G = S_G \circ F_0^G \circ S_G$ for each gadget. Differentiating at the 2-cycle points yields

$$J_1^A = DS_A\, J_0^A\, DS_A, \qquad J_1^B = DS_B\, J_0^B\, DS_B,$$

where the differentials of the phase-swap maps are the constant matrices

$$DS_A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad DS_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Lemma 6 (Half-step factorization)**.** *For each gadget $G \in \{A, B\}$, let $R_G = J_1^G J_0^G$ be the full return map with dominant eigenvalue μ_G ($\mu_A = \lambda$, $\mu_B = \kappa$). Let v_0^G be a corresponding eigenvector normalized so that its first entry (the p-component, in the chart coordinates of Section 3) is* 1*, and set $v_1^G = DS_G\, v_0^G$. Then, for $c_G := \sqrt{\mu_G} > 0$, we have*

$$J_0^G v_0^G = c_G v_1^G, \qquad J_1^G v_1^G = c_G v_0^G.$$

Proof. We give the argument for gadget A; gadget B is identical with DS_B, κ replacing DS_A, λ, once we verify that κ is a simple eigenvalue of R_B.

Conjugate return map. Define the *reverse return map* $R'_A = J_0^A J_1^A$. Using $J_1^A = DS_A J_0^A DS_A$ and $DS_A^2 = I$,

$$DS_A R_A DS_A = DS_A J_1^A J_0^A DS_A = (DS_A J_1^A DS_A)(DS_A J_0^A DS_A) = J_0^A J_1^A = R'_A.$$

Hence R'_A is conjugate to R_A and has the same eigenvalues λ, ν. Moreover,

$$R'_A v_1^A = R'_A(DS_A v_0^A) = DS_A R_A v_0^A = \lambda\, DS_A v_0^A = \lambda v_1^A,$$

so v_1^A is the λ-eigenvector of R'_A.

From $R_A v_0^A = \lambda v_0^A$ we get $J_1^A J_0^A v_0^A = \lambda v_0^A$. Therefore, with $w = J_0^A v_0^A$,

$$R'_A w = J_0^A(J_1^A w) = J_0^A(\lambda v_0^A) = \lambda J_0^A v_0^A = \lambda w.$$

Since $\lambda \neq \nu$, the λ-eigenspace of R'_A is one-dimensional. Because both w and v_1^A are λ-eigenvectors, there exists a scalar c_A such that $J_0^A v_0^A = c_A v_1^A$.

The second half-step identity and $c_A^2 = \lambda$. Using $J_1^A = DS_A J_0^A DS_A$, $v_1^A = DS_A v_0^A$, and $DS_A^2 = I$:

$$J_1^A v_1^A = DS_A J_0^A DS_A(DS_A v_0^A) = DS_A J_0^A v_0^A = DS_A(c_A v_1^A) = c_A v_0^A.$$

Composing the two half-steps gives

$$R_A v_0^A = J_1^A(c_A v_1^A) = c_A(c_A v_0^A) = c_A^2 v_0^A = \lambda v_0^A,$$

so, since $v_0^A \neq 0$, $c_A^2 = \lambda$.

Positivity from certified half-step p-multipliers. Because the p-component of v_1^A is exactly 1, the scalar c_A is the p-component of $J_0^A v_0^A$. The certificate (Item 9 of Appendix A.2) computes this p-component directly and proves

$$(J_0^A v_0^A)_p,\ (J_1^A v_1^A)_p \in [0.307395607438829821,\ 0.307395607492351897] \subset (0,1).$$

Hence $c_A > 0$. Since $c_A^2 = \lambda$, it follows that $c_A = \sqrt{\lambda}$.

Gadget B: simplicity of κ and positivity. As argued above, R_B has distinct eigenvalues λ, ν, κ. Here, κ is a simple eigenvalue of the block-triangular R_B, and the same argument applies. The certificate certifies

$$(J_0^B v_0^B)_p \in [0.451532647302682166,\ 0.451532650145556158],$$

$$(J_1^B v_1^B)_p \in [0.451532647288647226,\ 0.451532650159591153] \subset (0,1),$$

so $c_B > 0$, and therefore $c_B = \sqrt{\kappa}$. □

A-gadget eigenvectors and contraction

We now characterize eigenvectors for each gadget and verify the half-step contraction. Since $\lambda \neq \nu$, the eigenvalue λ of R_A is simple. Define θ_A as the unique scalar such that $R_A(1,\theta_A)^\top = \lambda(1,\theta_A)^\top$, and set $v_0^A = (1,\theta_A)$. The certificate (Item 8 of Appendix A.2) verifies

$$\theta_A \in [0.580405272518281104,\ 0.580405291461442915].$$

Set $v_1^A = DS_A v_0^A = (1, -\theta_A)$. By Lemma 6, for $c_A := \sqrt{\lambda}$, $J_0^A v_0^A = c_A v_1^A$, $J_1^A v_1^A = c_A v_0^A$. Moreover,

$$c_A \in [0.307395607465570497,\ 0.307395607465570659] < 1.$$

B-gadget eigenvectors and contraction

Since $\kappa \notin \{\lambda, \nu\}$, the eigenvalue κ of R_B is simple. Define $(\theta_{B,0}, \chi_B)$ as the unique pair such that

$$R_B(1, \theta_{B,0}, \chi_B)^\top = \kappa(1, \theta_{B,0}, \chi_B)^\top$$

and set $v_0^B = (1, \theta_{B,0}, \chi_B)$. The certificate (also Item 8 of Appendix A.2) verifies

$$\theta_{B,0} \in [0.198318090249174405,\ 0.198318111627417224], \tag{8.1}$$

$$\chi_B \in [1.01349886578531906,\ 1.01349888527878102]. \tag{8.2}$$

Set

$$v_1^B = DS_B v_0^B = (1, \theta_{B,1}, \chi_B),$$
$$\theta_{B,1} = -(\theta_{B,0} + \chi_B) \in [-1.21181699690619825,\ -1.21181695603449347].$$

By Lemma 6, for $c_B := \sqrt{\kappa}$, $J_0^B v_0^B = c_B v_1^B$, $J_1^B v_1^B = c_B v_0^B$. Moreover,

$$c_B \in [0.451532648724610500,\ 0.451532648724610611] \in (0, 1).$$

9 Explicit phase sectors and their certified invariance

We define explicit cone-like sectors around each phase of the 2-cycle and certify their forward invariance under the branch maps via rigorous interval arithmetic.

Review of interval arithmetic

We briefly discuss the framework. In *interval arithmetic* [5], every real quantity is enclosed by a closed interval $[\underline{x}, \overline{x}] \subset \mathbb{R}$ with $\underline{x} \le \overline{x}$, and each elementary operation $\circ \in \{+, -, \times, \div\}$ is replaced by an interval extension satisfying the *inclusion property*: whenever $x \in [\underline{x}, \overline{x}]$ and $y \in [\underline{y}, \overline{y}]$,

$$x \circ y \ \in\ [\underline{x}, \overline{x}] \circ [\underline{y}, \overline{y}].$$

For example, $[\underline{x}, \overline{x}] + [\underline{y}, \overline{y}] = [\underline{x} + \underline{y},\ \overline{x} + \overline{y}]$ and $[\underline{x}, \overline{x}] \times [\underline{y}, \overline{y}] = [\min S, \max S]$ where $S = \{\underline{x}\,\underline{y},\ \underline{x}\,\overline{y},\ \overline{x}\,\underline{y},\ \overline{x}\,\overline{y}\}$; division requires $0 \notin [\underline{y}, \overline{y}]$. By induction on the number of operations, evaluating any rational function f via interval arithmetic on an axis-aligned box $B \subset \mathbb{R}^n$—provided no intermediate divisor interval contains zero—yields an interval I satisfying $f(x) \in I$ for every $x \in B$; that is, $I \supseteq \{f(x) : x \in B\}$.

In our certificate, all interval endpoints are exact rational numbers, so every enclosure is justified. Also, Lemma 5 guarantees that all denominators arising from the branch maps are nonzero on the relevant domains, ensuring that no division by an interval containing zero occurs. This is the mechanism by which the branch maps, their Jacobians, and their Hessians are enclosed on the phase boxes defined below.

We consider two accuracy parameters r and ρ, where r bounds the principal coordinate (distance from the 2-cycle along the *dominant eigendirection*, which is the the direction of v_i^G) and ρ bounds the transverse slope. We set them as $r = 2.5 \times 10^{-8}$, $\rho = 10^{-3}$,

Below we will use standard notation for interval arithmetic, for instance, multiplying intervals with scalars as $a[b, c] = [ab, ac]$, for $a > 0$, $b \le c$. We emphasize that we always used closed intervals $[b, v]$ for interval arithmetic, and thus the notation (b, c) refers to a two-dimensional row vector and not an interval.

A-sectors

For each phase $i \in \{0,1\}$, the A-sector is the set

$$S_i^A = \Big\{ \hat{P}_i^A + x\, v_i^A + y\,(0,1) : 0 < x \le r,\ |y| \le \rho x \Big\}.$$

B-sectors

The B-sectors have two transverse directions:

$$S_i^B = \Big\{ \hat{P}_i^B + x\, v_i^B + y\,(0,1,0) + z\,(0,0,1) : 0 < x \le r,\ |y| \le \rho x,\ |z| \le \rho x \Big\}.$$

Recalling the certified interval (8.2), $\chi_B > 1.013$, so $\chi_B - \rho > 1.013 - 10^{-3} > 1$. Hence, every point of S_i^B has $e > 0$, so $S_i^B \subset \Sigma_B^\circ$.

Lemma 7 (Exact first-column structure in the adapted bases)**.** *Define the adapted basis matrices*

$$B_0^A = [v_0^A, (0,1)], \qquad B_1^A = [v_1^A, (0,1)],$$
$$B_0^B = [v_0^B, (0,1,0), (0,0,1)], \qquad B_1^B = [v_1^B, (0,1,0), (0,0,1)].$$

Then, for $i \in \{0,1\}$ (where $i \oplus 1$ denotes addition modulo 2),

$$(B_{i\oplus 1}^A)^{-1} J_i^A B_i^A = \begin{pmatrix} c_A & * \\ 0 & * \end{pmatrix},$$

and

$$(B_{i\oplus 1}^B)^{-1} J_i^B B_i^B = \begin{pmatrix} c_B & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}.$$

Proof. The first column of B_i^A is v_i^A. By Lemma 6, $J_i^A v_i^A = c_A v_{i\oplus 1}^A$. Since $v_{i\oplus 1}^A$ is the first column of $B_{i\oplus 1}^A$, left-multiplying by $(B_{i\oplus 1}^A)^{-1}$ gives first coordinate c_A and zero transverse coordinate. That is exactly the displayed matrix form. The B-case is identical, using $J_i^B v_i^B = c_B v_{i\oplus 1}^B$. □

For each sector S_i^G, a *phase box* is a coordinate-aligned rectangular box (Cartesian product of closed intervals in the chart coordinates) that contains the closure of S_i^G; the certificate uses phase boxes because interval arithmetic operates on such axis-aligned rectangles. The certificate (Item 10 of Appendix A.2) computes interval Jacobians in these exact bases and second-derivative bounds on explicit rational phase boxes covering the sectors.

The B-phase boxes extend to $e = 0$ (their e-coordinate ranges over $[0, k_{Be} r]$); since $Q_B|_{e=0} = Q_A$ and $R_B|_{e=0} = R_A$. Lemma 5 (applied to the A-gadget at $e = 0$ and to the B-gadget for $e > 0$) ensures that all denominators are nonzero on the full closed phase boxes, so every interval-arithmetic division is well-defined. By Lemma 7, the lower entries in the first transformed column vanish exactly, so the sector inequalities only use the remaining coefficients and the certified Hessian remainder terms. The resulting certified half-step bounds are listed below:

A-sector certified bounds

In the sector coordinates, the ratio x'/x measures the contraction factor along the dominant eigendirection (and should be close to the half-step eigenvalue $c_A \approx 0.3074$), while $|y'|/x'$ measures the transverse drift relative to the cone boundary ρ.

For both phases $i = 0, 1$,

$$\frac{x'}{x} \in [0.307394561088771942,\ 0.307396653842409774],$$
$$\frac{|y'|}{x'} \le 1.10321761561807761 \times 10^{-4} < \rho.$$

(For phase 0 the certified bound is slightly smaller, $1.10321759830186699 \times 10^{-4}$.)

B-sector certified bounds

At phase 0:

$$\frac{x'}{x} \in [0.451388989838190632,\ 0.451676307610047724],$$
$$\frac{\max(|y'|, |z'|)}{x'} \le 6.82589234315899433 \times 10^{-4} < \rho.$$

At phase 1:

$$\frac{x'}{x} \in [0.451389955166172293,\ 0.451675342282066063],$$
$$\frac{\max(|y'|, |z'|)}{x'} \le 6.84951234683088803 \times 10^{-4} < \rho.$$

The A-gadget bounds above are uniform over both phases (the phase-0 bound on $|y'|/x'$ is slightly tighter, as noted in the parenthetical), so they are reported together. The B-gadget's 3-dimensional dynamics make the two phases slightly asymmetric, so we report them separately above.

Therefore

$$F_i^A(S_i^A) \subset S_{i\oplus 1}^A, \qquad F_i^B(S_i^B) \subset S_{i\oplus 1}^B.$$

Define the transverse ratios $\tau_m^A = |y_m|/x_m^A$ and $\tau_m^B = \max(|y_m|, |z_m|)/x_m^B$. The same computation yields explicit transverse-ratio recurrences, which will be needed for the asymptotic analysis in Section 12:

$$\begin{aligned}
\tau_{m+1}^A &\le \alpha_A \tau_m^A + \beta_A x_m^A, \\
&\qquad \alpha_A \le 0.110098192661922619, \quad \beta_A \le 8.94275599540566418, \\
\tau_{m+1}^B &\le \alpha_B \tau_m^B + \beta_B x_m^B, \\
&\qquad \alpha_B \le 0.684664374851195431, \quad \beta_B \le 12.8511069222196774,
\end{aligned}$$

Here m indexes the successive bursts won by each respective gadget. Since $\alpha_A, \alpha_B < 1$ and the principal coordinates are geometrically summable, the transverse-ratio series $\sum_m \tau_m^A$ and $\sum_m \tau_m^B$ converge; the full argument is given in Section 12.

10 Local branch words and whole-burst locking

Within the invariant sectors, we must verify two properties: that the column-selection sequence within each gadget matches the expected half-word (H_0 or H_1), and that once a gadget wins the first round of a 5-step burst, it wins all five rounds. Both are established by certified gap bounds.

The certificate (Items 11–12 of Appendix A.2) checks that at each step of each half-word on the full phase boxes, the corresponding column has the largest edge among the four columns of the active gadget. The minimum margin is

$$g_{\mathrm{br}}^{\mathrm{box}} > 0.0250983508506417014.$$

It also certifies that at steps $2, 3, 4, 5$ of each half-word the selected edge exceeds $\mu_* = 2p_* - 1$ by at least

$$\Delta_{\mathrm{box}} > 0.0758691457701749199.$$

The largest possible first-edge surplus on any phase box is only

$$\delta_{\mathrm{first}}^{\max} < 1.05687287632489371 \times 10^{-7}.$$

Consequently, at steps $2, 3, 4, 5$ of any burst the active gadget's selected edge exceeds the best edge of the frozen gadget by at least $\Delta_{\mathrm{box}} - \delta_{\mathrm{first}}^{\max}$ (which we denote as the *whole-burst lock margin*):

$$\Delta_{\mathrm{box}} - \delta_{\mathrm{first}}^{\max} > 0.0758690400828872874.$$

These bounds have three immediate consequences. First, the column-selection sequence within the A-gadget is exactly H_0 on S_0^A and exactly H_1 on S_1^A; likewise for the B-gadget on S_0^B and S_1^B. Second, and more importantly, once a gadget wins the first round of a burst, it stays globally ahead for the remaining four rounds of that burst. To see this, observe that during an A-burst the B-state does not move at all, so the best B-correlation throughout the burst is bounded by its first-round best edge; the same holds with A and B reversed. Since the strict margin $\Delta_{\mathrm{box}} - \delta_{\mathrm{first}}^{\max} > 0$ separates the selected edge from the first-edge surplus, the first-round winner necessarily keeps winning for rounds $2, 3, 4, 5$.

11 Product burst dynamics

Let $i_n \in \{0, 1\}$ and $j_n \in \{0, 1\}$ denote the A-phase and B-phase at burst boundary n, respectively. Because the starting point lies in $S_0^A \times S_0^B$ (Section 14), the certified sector invariance $F_i^G(S_i^G) \subset S_{i\oplus 1}^G$ from Section 9 and the whole-burst lock from Section 10 imply by induction that $\hat{a}_n \in S_{i_n}^A$ and $\hat{b}_n \in S_{j_n}^B$ for every burst boundary n.

Let $W_n = a_n \otimes b_n \in \mathcal{P}$ be the burst-boundary states, where $a_n \in \Delta^3$ and $b_n \in \Delta^4$ are the simplex marginals. Write the corresponding chart coordinates as

$$\begin{aligned} \hat{a}_n &= \hat{P}_{i_n}^A + x_n^A v_{i_n}^A + y_n^A(0, 1), \\ \hat{b}_n &= \hat{P}_{j_n}^B + x_n^B v_{j_n}^B + y_n^B(0, 1, 0) + z_n^B(0, 0, 1), \end{aligned} \tag{11.1}$$

with $\hat{a}_n \in S_{i_n}^A$ and $\hat{b}_n \in S_{j_n}^B$. For brevity write $x_n := x_n^A$, $s_n := x_n^B$.

At the start of a burst, the first selected local edge in either gadget is exactly $2p-1 = \mu_* + 2(p-p_*)$. Since in the sector coordinates $p - p_* = x$, the first global round chooses

- A if $x_n > s_n$;
- B if $s_n > x_n$;
- A on equality, by smallest-index tie-breaking, because the A-columns have smaller global indices than the B-columns.

Because of the whole-burst lock margin from Section 10, that first-round winner then wins the entire 5-step burst.

Half-step contraction bounds

The certified sector bounds from Section 9 yield the following contraction bounds. If A wins burst n, then $x_{n+1} \in [m_A, M_A]\, x_n, s_{n+1} = s_n$, with

$$m_A = 0.307394561088771942, \qquad M_A = 0.307396653842409774.$$

If B wins burst n, then $s_{n+1} \in [m_B, M_B]\, s_n, x_{n+1} = x_n$, with

$$m_B = 0.451388989838190632, \qquad M_B = 0.451676307610047724.$$

Writing $r_n = x_n/s_n$, the contraction bounds above translate to: if A wins burst n, then $r_{n+1} \in [m_A r_n,\, M_A r_n]$; if B wins, then $r_{n+1} \in [r_n/M_B,\, r_n/m_B]$.

Since A wins precisely when $r_n \geq 1$ and B wins when $r_n < 1$, the ratio r_n has a multiplicative dynamics that we can summarize as follows.

Lemma 8 (Burst-ratio dynamics and run-length bounds)**.** *For all sufficiently large n, one has $r_n \in [m_A,\, 1/m_B]$. Moreover:*

(i) every A-run has length exactly 1;

(ii) every B-run has length at most 2.

Consequently, every block of three consecutive bursts contains at least one A-burst and at least one B-burst, and $x_n + s_n \to 0$.

Proof. **Entry into the compact interval.** After a B-burst at step n (so $r_n < 1$), we have $r_{n+1} \leq r_n/m_B < 1/m_B$. After an A-burst at step n (so $r_n \geq 1$), we have $r_{n+1} \geq m_A r_n \geq m_A$. Once the trajectory has experienced at least one A-burst and one B-burst, r_n enters and remains in $[m_A, 1/m_B]$. Both burst types must eventually occur: if only A-bursts occurred, then $r_n \leq M_A^n r_0 \to 0$ (since $M_A < 1$), eventually forcing $r_n < 1$ and hence a B-burst; symmetrically, if only B-bursts occurred, then $r_n \geq r_0/M_B^n \to \infty$ (since $M_B < 1$), eventually forcing $r_n \geq 1$ and hence an A-burst.

A-runs have length 1. Suppose A wins at step n, so $r_n \geq 1$, and that $r_n \leq 1/m_B$ (since we are in the compact interval). Then,

$$r_{n+1} \leq M_A r_n \leq \frac{M_A}{m_B} < 1,$$

where the last inequality is the certified bound $M_A/m_B < 1$. Hence B wins at step $n+1$, and every A-run has length exactly 1.

B-runs have length at most 2. Suppose B wins at step n, so $r_n < 1$, and that $r_n \geq m_A$. After one B-burst,

$$r_{n+1} \geq \frac{r_n}{M_B} \geq \frac{m_A}{M_B}.$$

If $r_{n+1} \geq 1$, the B-run ends after one step. If $r_{n+1} < 1$, then B wins again, and

$$r_{n+2} \geq \frac{r_{n+1}}{M_B} \geq \frac{m_A}{M_B^2} > 1,$$

where the last inequality is the certified bound $m_A/M_B^2 > 1$. Hence A wins at step $n+2$, and every B-run has length at most 2.

Exponential convergence. By Steps 2 and 3, every block of three consecutive bursts contains at least one A-burst and at least one B-burst, so

$$x_{n+3} + s_{n+3} \le \theta\,(x_n + s_n), \qquad \theta = \max(M_A, M_B) < 1.$$

This finishes the proof. □

12 Local asymptotics on the two gadgets

We now establish that the principal coordinates along each gadget's dominant eigendirection contract at rates asymptotically governed by $c_A = \sqrt{\lambda}$ and $c_B = \sqrt{\kappa}$, with summable relative errors. Let $m \mapsto \hat{a}^{(m)}$ be the chart coordinates of the local A-state immediately before the mth A-burst, and $m \mapsto \hat{b}^{(m)}$ the chart coordinates of the local B-state immediately before the mth B-burst. Write

$$x_m^A = p(\hat{a}^{(m)}) - p_*, \qquad x_m^B = p(\hat{b}^{(m)}) - p_*.$$

Because the exact basis vectors from Section 8 satisfy $J_i^A v_i^A = c_A v_{i\oplus 1}^A$, $J_i^B v_i^B = c_B v_{i\oplus 1}^B$, the certified Taylor bounds from Section 9 give

$$\begin{aligned} x_{m+1}^A &= c_A x_m^A\big(1+\varepsilon_m^A\big), & |\varepsilon_m^A| &\le \Gamma_A \tau_m^A + \Delta_A x_m^A, \\ x_{m+1}^B &= c_B x_m^B\big(1+\varepsilon_m^B\big), & |\varepsilon_m^B| &\le \Gamma_B \tau_m^B + \Delta_B x_m^B, \end{aligned}$$

with certified constants

$$\begin{aligned} \Gamma_A &\le 0.003273115711042833, & \Delta_A &\le 5.232171784223787, \\ \Gamma_B &\le 0.31803397447249865, & \Delta_B &\le 8.463704731868965, \end{aligned}$$

and therefore uniform relative-error bounds

$$\begin{aligned} \Gamma_A \tau + \Delta_A x &\le 0.00000340392000564842791 && \text{throughout the A sectors,} \\ \Gamma_B \tau + \Delta_B x &\le 0.00031824556709079535 0 && \text{throughout the B sectors.} \end{aligned}$$

In particular, the relative errors are everywhere < 0.01.

Since $x_{m+1}^A \le M_A x_m^A$ and $x_{m+1}^B \le M_B x_m^B$, both local x-sequences are summable. Iterating the transverse recurrence $\tau_{m+1} \le \alpha\tau_m + \beta x_m$ gives $\tau_m \le \alpha^m \tau_0 + \beta \sum_{k=0}^{m-1} \alpha^{m-1-k} x_k$. Summing over m and exchanging the order of summation,

$$\sum_m \tau_m \le \frac{\tau_0}{1-\alpha} + \frac{\beta}{1-\alpha} \sum_k x_k < \infty.$$

In particular, both transverse-ratio series $\sum_m \tau_m^A$ and $\sum_m \tau_m^B$ converge, and consequently

$$\sum_m |\varepsilon_m^A| < \infty, \qquad \sum_m |\varepsilon_m^B| < \infty.$$

Since $|\varepsilon_m^A|, |\varepsilon_m^B| < 0.01$ throughout, the bound $|\log(1+u)| \le 2|u|$ applies, giving

$$\begin{aligned} \log x_m^A &= \log x_0^A + m \log c_A + O(1), \\ \log x_m^B &= \log x_0^B + m \log c_B + O(1). \end{aligned}$$

Let $N_A(n)$ denote the number of A-bursts among bursts $0, \ldots, n-1$, and let $N_B(n) = n - N_A(n)$. At burst boundary n, the A-state has been updated exactly $N_A(n)$ times and the B-state exactly $N_B(n)$ times, so

$$\log x_n = \log x_0 + N_A(n) \log c_A + O(1),$$
$$\log s_n = \log s_0 + N_B(n) \log c_B + O(1).$$

Because $r_n = x_n/s_n$ stays in the compact interval $[m_A, 1/m_B]$, the logarithmic difference is bounded:

$$N_A(n) \log c_A - N_B(n) \log c_B = O(1).$$

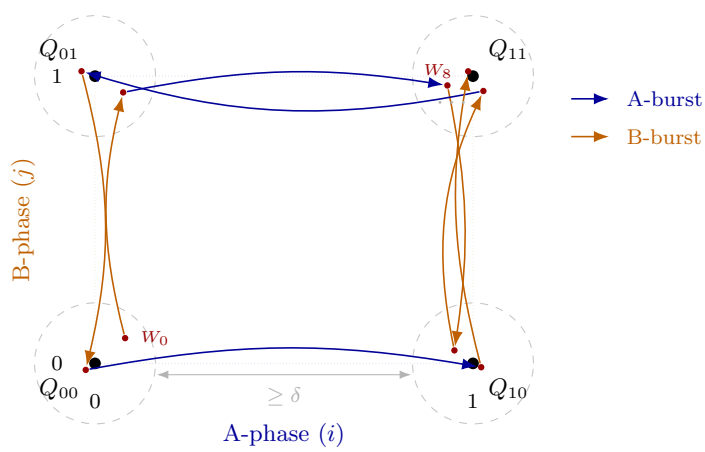
Since $N_A(n) + N_B(n) = n$, it follows that

$$\frac{N_A(n)}{n} \to \frac{-\log c_B}{-\log c_A - \log c_B} = \frac{-\log \kappa}{-\log \lambda - \log \kappa}.$$

Because $\log \lambda / \log \kappa \notin \mathbb{Q}$ (Section 7), this limiting frequency is irrational. Since every eventually periodic binary sequence has rational limiting frequency, the burst-winner sequence is not eventually periodic.

13 No finite-cycle convergence

We now combine the preceding results to show that the AdaBoost orbit on M_0 does not converge to any finite cycle. The argument proceeds by contradiction: assuming cycle convergence, we deduce that the burst-winner sequence is eventually periodic, contradicting its irrational asymptotic frequency.



$$\frac{N_A(n)}{n} \to \frac{-\log \kappa}{-\log \lambda - \log \kappa} \notin \mathbb{Q} \implies \text{burst-winner sequence is aperiodic}$$

Figure 2: Schematic of the no-finite-cycle argument. The four points $Q_{ij} = P_i^A \otimes P_j^B$ sit at the vertices of a phase square in $\mathcal{P}$, pairwise separated by at least $\delta > 0$. Each A-burst (blue) toggles the A-phase i, and each B-burst (orange) toggles the B-phase j. The burst-boundary states W_n (red dots) converge toward these vertices as $\|W_n - Q_{i_n j_n}\| \to 0$, but the irrational limiting frequency of A-bursts precludes eventual periodicity of the burst-winner sequence and hence convergence of the AdaBoost orbit to any finite cycle.

Recalling (6.1), (6.2), define $Q_{ij} = P_i^A \otimes P_j^B$, $i, j \in \{0, 1\}$. These four points are distinct: the tensor product $P_i^A \otimes P_j^B$ determines its marginals by summation, and $P_0^A \neq P_1^A$, $P_0^B \neq P_1^B$ (their 3rd and 4th coordinates are swapped, with $d_* \neq \frac{1}{4}$). Since $x_n, s_n \to 0$, the burst-boundary states satisfy $\|W_n - Q_{i_n j_n}\| \to 0$. Indeed, first recall that $W_n = a_n \otimes b_n \in \mathcal{P}$. Then, mapping the sector parametrization (11.1) to the affine chart gives $\|a_n - P_{i_n}^A\|_1 = O(x_n)$ and $\|b_n - P_{j_n}^B\|_1 = O(s_n)$, while for distributions $\|a \otimes b - a' \otimes b'\|_1 \leq \|a - a'\|_1 + \|b - b'\|_1$; the claim then follows from $x_n + s_n \to 0$ (Lemma 8).

Suppose for contradiction that the full AdaBoost trajectory converges to a cycle of length ℓ, meaning $D_{k\ell+b} \to \widetilde{D}_b$, $(k \to \infty)$ for each $b \in \{0, \ldots, \ell - 1\}$. Each burst has exactly 5 update steps. Recall from Section 2 that D_t denotes the state at time t; following the convention of [7], the sequence begins at $t = 1$, so D_1 is the initial distribution. Since $W_0 = D_1$ and each burst comprises 5 update steps, the state after n complete bursts is $W_n = D_{5n+1}$. Set $L = \ell / \gcd(5, \ell)$. Then for each residue $r \in \{0, \ldots, L - 1\}$, the subsequence W_{kL+r} converges to a limit $\widetilde{W}_r$ as $k \to \infty$.

Let

$$\delta = \min\{\|Q - Q'\| : Q, Q' \in \{Q_{00}, Q_{01}, Q_{10}, Q_{11}\},\ Q \neq Q'\} > 0.$$

Fix a residue r, and abbreviate $Q_k := Q_{i_{kL+r}, j_{kL+r}}$. Choose $\varepsilon = \delta/4$. Because $W_{kL+r} \to \widetilde{W}_r$ and $\|W_n - Q_{i_n j_n}\| \to 0$, there is K_r such that for all $k \geq K_r$,

$$\|W_{kL+r} - \widetilde{W}_r\| < \varepsilon, \qquad \|W_{kL+r} - Q_k\| < \varepsilon.$$

Now take any $k, k' \geq K_r$. Then

$$\|Q_k - Q_{k'}\| \leq \|Q_k - W_{kL+r}\| + \|W_{kL+r} - \widetilde{W}_r\| + \|\widetilde{W}_r - W_{k'L+r}\| + \|W_{k'L+r} - Q_{k'}\| < 4\varepsilon = \delta.$$

But distinct points among $\{Q_{00}, Q_{01}, Q_{10}, Q_{11}\}$ are at least δ apart. Therefore $Q_k = Q_{k'}$ for all $k, k' \geq K_r$. So the phase label (i_{kL+r}, j_{kL+r}) is eventually constant on each residue class r.

Hence the phase-label sequence (i_n, j_n) is eventually periodic. Since each burst toggles exactly one coordinate of (i_n, j_n)—the active gadget's phase, by Section 9; the inactive gadget is unchanged by Lemma 4—the burst-winner sequence is determined by the transitions $(i_n, j_n) \to (i_{n+1}, j_{n+1})$ and is therefore also eventually periodic. That contradicts Section 12, which showed that the burst-winner sequence has an irrational limiting frequency and therefore cannot be eventually periodic. Therefore the AdaBoost orbit on M_0 does not converge to any finite cycle.

14 Explicit rational start and explicit uniform-start counterexample

It remains to exhibit a concrete starting point inside the invariant sectors and to convert it into a uniform-start counterexample. We begin by taking

$$A = (137116847818847,\ 362883152181153,\ 242004611239991,\ 257995388760009),$$
$$B = (137116857552466,\ 362883142447534,\ 242004593130510,\ 257995386869490,\ 20000000).$$

Since simple arithmetic shows that $\sum_i A_i = \sum_j B_j = 10^{15}$, set $a_0 = \frac{A}{10^{15}}$, $b_0 = \frac{B}{10^{15}}$, $W_0 = a_0 \otimes b_0$. To apply the sector-based analysis of Sections 9–13, we need to verify that this starting point lies inside the invariant sectors.

The certificate (Item 13 of Appendix A.2) proves that the chart images satisfy $\hat{a}_0 \in S_0^A$, $\hat{b}_0 \in S_0^B$. The starting point is decomposed in sector coordinates: we report the principal coordinate x

(distance from the 2-cycle along the dominant eigendirection) and the transverse ratios (deviation from the eigendirection, normalized by x), confirming that they satisfy the sector constraint $|\text{transverse ratio}| \le \rho$.

More explicitly: $x_A \in [9.999999 \times 10^{-9},\ 1.0000001 \times 10^{-8}]$, with A-sector residual in

$$[-2.18994134061382760 \times 10^{-4},\ 2.19798676726244218 \times 10^{-4}],$$

and $x_B \in [1.9733618 \times 10^{-8},\ 1.9733620 \times 10^{-8}]$, with B d-residual in

$$[-1.10962941057972692 \times 10^{-4},\ 1.11368115987132873 \times 10^{-4}],$$

and B e-residual in

$$[-9.48895873525338312 \times 10^{-8},\ 2.73218648306331209 \times 10^{-8}],$$

all strictly inside the sector slope bound $\rho = 10^{-3}$. Note that $x_A \approx 10^{-8} < 1.97 \times 10^{-8} \approx x_B$, so $r_0 = x_A/x_B < 1$ and the first burst is a B-burst.

Lemma 9 (Row duplication and class aggregation)**.** *Let M' be obtained from a matrix M by duplicating rows. Partition the rows of M' into duplicate classes, one class for each row of M. Suppose the initial distribution on M' is constant on each duplicate class.*

Then:

1. *the AdaBoost orbit on M' remains constant on each duplicate class for all time;*
2. *the class-summed weights evolve exactly as exhaustive AdaBoost on M started from the corresponding aggregated distribution;*
3. *$\gamma(M') = \gamma(M)$.*

Proof. Identical rows have the same sign in every column, so at every AdaBoost step they are multiplied by the same factor $1/(1 + r_t h_i)$. Hence equality within each duplicate class is preserved forever.

Let $\bar{D}_t$ be the distribution obtained by summing D'_t over each duplicate class. Because every row in a given class has the same entries in every column, the edge vector of D'_t equals the edge vector of $\bar{D}_t$. Therefore the selected column is the same for D'_t and $\bar{D}_t$, and the class sums satisfy exactly the same update formula as exhaustive AdaBoost on M.

For the margin, every distribution on M' aggregates to a distribution on M with the same edge vector, so $\gamma(M') \ge \gamma(M)$. Conversely, any distribution on M lifts to a class-constant distribution on M' with the same edge vector, so $\gamma(M') \le \gamma(M)$. Hence $\gamma(M') = \gamma(M)$. □

Now construct $\widetilde{M}$ by replacing each row (i, j) of M_0 by $A_i B_j$ identical copies. Since $\sum_i A_i \sum_j B_j = 10^{30}$, the exact uniform distribution on $\widetilde{M}$ aggregates to W_0. By Lemma 9, the class-aggregated weights of the uniform-start AdaBoost orbit on $\widetilde{M}$ correspond exactly to the weights on the W_0-started orbit on M_0. Since the latter does not converge to a finite cycle, neither does the former.

By Lemma 9 and Section 4, $\gamma(\widetilde{M}) = \gamma(M_0) = \frac{1}{5}$. This completes the proof of Theorem 1. □

Proof of Corollary 2. Set $\widetilde{M}' = [\widetilde{M} \mid -\widetilde{M}]$, where the columns of $-\widetilde{M}$ are appended after those of $\widetilde{M}$. By construction, the column set of $\widetilde{M}'$ is closed under negation.

We verify that no negated column is ever selected, so that the dynamics on $\widetilde{M}'$ are identical to those on $\widetilde{M}$. By Lemma 9, the edge vectors on $\widetilde{M}$ equal those on M_0 at every step, so it suffices to

verify on M_0 that the selected column edge strictly exceeds every negated column edge at every step. The certificate does so by exact interval arithmetic on the full sector boxes (items 15–16 of Appendix A.2). During any burst, two types of negated columns could compete: those from the active gadget and those from the frozen (non-active) gadget.

For the frozen gadget, Lemma 3 implies its column edges do not change during the burst; the certificate encloses all frozen-gadget column edges on each sector box. For the active gadget, at each of the five steps within each burst word on each sector box, the certificate (Item 15 of Appendix A.2) encloses all column edges. At every step of every burst/frozen-phase combination, the certificate (Item 15 of Appendix A.2) verifies that the selected edge strictly exceeds every negated column edge. The minimum certified gap over all eight burst/frozen-phase combinations and all five steps is

$$g_{\text{neg}}^{\text{box}} \geq 0.274233539263122735 > 0.$$

Hence no negated column is ever in the argmax set, so the dynamics on $\widetilde{M}'$ are identical to those on $\widetilde{M}$, which by the Theorem does not converge to any finite cycle.

For the margin, the minimax distributions $u_*^A = (1/5, 2/5, 1/5, 1/5)$ on L_A and $u_*^B = (1/5, 2/5, 1/5, 1/5, 0)$ on L_B each give all four column edges equal to exactly $1/5$. Hence every negated column edge is $-1/5$, and $\gamma(\widetilde{M}') = \gamma(\widetilde{M}) = 1/5$. □

A Appendix

A.1 Additional results

Lemma 10 (Balancing after update [8])**.** *A single AdaBoost update on a selected column $h \in \{\pm 1\}^m$ sends the total post-update mass on the $h = +1$ rows and on the $h = -1$ rows to exactly $1/2$ each.*

Proof. If $\mu = \sum_i w_i h_i$, then the exhaustive AdaBoost update is $w_i' = \frac{w_i}{1+\mu h_i}$. Therefore

$$\sum_{h_i=+1} w_i' = \frac{\sum_{h_i=+1} w_i}{1+\mu} = \frac{(1+\mu)/2}{1+\mu} = \frac{1}{2},$$

and similarly $\sum_{h_i=-1} w_i' = 1/2$. □

A.2 What the certificate proves

The certificate proves all of the following finite assertions.

1. The explicit branch maps $F_0^A, F_1^A, F_0^B, F_1^B$ are derived symbolically from H_0, H_1, and the displayed numerator/denominator polynomials $(P_A, Q_A, D_A, R_A, P_B, Q_B, D_B, R_B, E_B)$ are verified to match the derived rational functions term-for-term.

2. $F_0^B|_{e=0} = F_0^A$, $F_1^B|_{e=0} = F_1^A$, and both B third-coordinate numerators are exactly divisible by e.

3. The 2-cycle equations reduce to $N_1 = N_2 = 0$; the displayed polynomials N_1, N_2 are verified to match the derived expressions term-for-term. Elimination gives the exact degree-7 polynomial f, and the root interval for p_* is unique.

4. The exact polynomial formula for d_* is correct modulo f.

5. The exact elimination polynomial for λ is the irreducible degree-14 polynomial above.

6. The exact elimination polynomial for κ is the irreducible degree-7 polynomial above.
7. The isolating intervals for λ, ν, κ each contain exactly one root.
8. The exact dominant half-step directions v_i^A, v_i^B are certified by explicit rational interval enclosures of $\theta_A, \theta_{B,0}, \chi_B, \theta_{B,1}$.
9. The half-step contractions c_A, c_B are certified and satisfy $c_A < 1$, $c_B < 1$.
10. The explicit sectors S_i^A, S_i^B with $r = 2.5 \times 10^{-8}$ and $\rho = 10^{-3}$ are certified by interval Jacobians and interval Hessian bounds, using the exact first-column structure from Lemma 7.
11. The local branch words H_0, H_1 are certified on the full phase boxes covering those sectors.
12. The whole-burst lock margin is certified.
13. The explicit rational starting point $W_0 = a_0 \otimes b_0$ is certified: its chart image satisfies $\hat{a}_0 \in S_0^A$, $\hat{b}_0 \in S_0^B$.
14. No floating-point number is used in any proof assertion. Floating-point/decimal conversion is used only for human-readable printing of already-certified rational intervals.
15. No negated column is ever selected on $[\widetilde{M} \mid -\widetilde{M}]$: the certificate computes, by exact interval arithmetic on the full sector boxes covering all eight burst/frozen-phase combinations and all five burst steps, interval enclosures of every column edge at every step, and verifies that the selected edge strictly exceeds every negated column edge. The certified minimum gap is $g_{\text{neg}}^{\text{box}} \geq 0.274233539263122735 > 0$.
16. The margin $\gamma([\widetilde{M} \mid -\widetilde{M}]) = 1/5$ is certified: the explicit minimax distributions $u_*^A = (1/5, 2/5, 1/5, 1/5)$ and $u_*^B = (1/5, 2/5, 1/5, 1/5, 0)$ are verified to give all column edges exactly $1/5$ on L_A and L_B respectively, so every negated column edge is $-1/5$.

In the certificate, items 1–14 are verified by `adaboost_cert_core.py`, and items 15–16 are verified by `adaboost_cert_negcols.py`. Running `adaboost_cert_full.py` executes both scripts and prints SHA-256 digests for reproducibility.

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